LIRUI WANG

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EDUCATION

University of Washington, Seattle, WA *Sep 2016 - Mar 2021*
Bachelor of Science, Electrical Engineering, (Honors), College of Engineering
Bachelor of Science, Computer Science, (Honors), College of Arts and Sciences
Bachelor of Science, Mathematics, (Minor), College of Arts and Sciences
Master of Science, Computer Science, College of Arts and Sciences
GPA: 3.96/4.00, *summa cum laude*, Advisor: Prof. Dieter Fox

RESEARCH PROJECTS

6D Instance Grasping with Point Cloud *Jan 2020 - Sep 2020*
- Augmented reinforcement learning with demonstrations and goal auxiliary tasks, for learning closed-loop control policies in 6D robotic grasping using point clouds
- Showed over 90% success rates on grasping various ShapeNet objects and YCB objects in simulation.

Joint Motion and Grasp Planning *March 2019 - Dec 2019*
- Proposed an optimization-based integrated planner that optimizes manipulation trajectory with online grasp synthesis and selection.
- Demonstrated robust and efficient motion plans for grasping in cluttered scenes in the real world and outperformed several baselines.

Pose Tracking with the Deep Iterative Matching Network *June 2018 - March 2019*
- Evaluated DeepIM on YCB dataset in depth and achieved robust results trained on synthetic data by improving network training stability, pose distribution and occlusion generation.
- Extended DeepIM to achieve state-of-the-art pose tracking accuracy and built real-world applications.

Unsupervised Flow and Depth Methods Study *March 2018 - June 2018*
- Investigated deep learning methods for 3D geometry tasks such as depths, flow, and egomotion.
- Explored unsupervised structures and ideas such as SfMlearner and MonoDepth and analyzed them extensively on a static scene manipulation setting.

Baxter Hand-Eye Coordination *Aug 2017 - March 2018*
- Developed software for the Baxter robot to grasp and manipulate daily objects detected by PoseCNN.
- Studied and implemented camera models, hand-eye calibrations, robot kinematics, renderer, and various ROS packages for tracking and control.

WORK EXPERIENCE

Seattle, WA *Sep 2020 - Dec 2020*
Deep Learning Teaching Assistant, UW CSE 490

Seattle, WA *June 2020 - Sep 2020*
Applied Research Intern, NVIDIA
- Investigated model-based RL for autonomous driving with a learned traffic prediction model.

- Trained actor critic on the black-box simulator to learn avoiding cut-in behavior and apply the value function for long-horizon motion planner.

Seattle, WA *Jan 2020 - Mar 2020*
Artificial Intelligence Teaching Assistant, UW CSE 473

Seattle, WA *Sep 2019 - Jan 2020*
Computer Vision Teaching Assistant, UW CSE 455

Shenzhen, China *Jun 2017 - Aug 2017*
Robotics Engineer Intern, DJI Technology

- Implemented functions for drones such as take-off, tag detection, and cooperate with ground vehicle.
- Applied SLAM to mapping and localization, and built a state machine to generate high-level plans.

LEADERSHIP AND INVOLVEMENT

Seattle, WA Sep 2016 - Feb 2018
Computer Vision Lead *Advanced Robotics at the University of Washington*

- Managed the a team of 10 students at UW to design computer vision algorithms for an autonomous robot battle, including vehicles with onboard NVIDIA TX1 and drones with DJI Manifold with Guidance, in the RoboMasters competition hosted by DJI.
- Performed system integration of feedback controls, navigation systems and serial communication.

MISCELLANEOUS PROJECTS

- **Spatial Audio Processing in the SoundScape System**: Designed a sound processing pipeline from scratch for modelling the acoustics of a train.
- **Sheet Music Editor**: Created a demo app that can record sound pieces and translate into sheet music display UI using DTFT, and it was awarded 1st place out of 35 teams in Seattle.
- **Embedded Drowsiness Detection System**: Built an embedded system on Raspberry Pi that that simulates a real-time drowsiness detection system based on human face detection.
- **Survey and Empirical Evaluation of Attacks and Defense for Robust Classification**: Investigated example reweighting and adversarial training techniques to robustify a fossil classification task and surveyed recent papers on attacks and defense of adversarial examples for neural networks.
- **Continuous Trajectory Optimization with Machine Learning Techniques**: Summarized the formulation of trajectory optimization as functional gradient descent using Reproducing Kernel Hilbert Space and as probabilistic inference using Gaussian Process.
- **Microsoft Server Mover Project**: Built an autonomous mobile manipulator, a UR5 arm mounted on a Server Lift, to pull out servers from server rack and transport it to repair.

HONORS

- Annual Deans List, University of Washington, *2016 - 2020*
- Tau Beta Pi Scholarship *2017 - 2019*
- Lawrence & Lucille Frey Endowed Electrical Engineering Scholarship *2017 - 2018*
- Patricia G. Lynch and Theodora & Eugene Russell Memorial Computer Science Scholarship *2018 - 2019*

PUBLICATIONS

- **Goal-Auxiliary Actor-Critic for 6D Robotic Grasping with Point Clouds**
  Lirui Wang, Yu Xiang, Dieter Fox
  Under Review
- **Manipulation Trajectory Optimization with Online Grasp Synthesis and Selection**
  Lirui Wang, Yu Xiang, Dieter Fox
  In Proceedings of Robotics: Science and Systems (RSS 2020)

SELECTED COURSES

- **Computer Science**: Data Structure and Parallelism, Algorithms, Computer Vision, Artificial Intelligence, Deep Learning, Modern Algorithm Toolbox, Graphics, Robotics, Robustness in Machine Learning (Grad), Interactive Learning (Grad), Robotics (Grad), Reinforcement Learning
- **Electrical Engineering**: Digital Signal Processing and Filtering, Control Theory, Embedded System, Discrete Time Linear System, Systems, Controls, and Robotics Capstone, Optimization and Learning for Control (Grad)
- **Math**: Multivariable Calculus, Probability and Statistics, Numerical Analysis, Linear Analysis, Topology, Convex Optimization and Analysis (Grad), High Dimensional Statistics and Statistical Learning (Grad), Statistical Learning: Modeling, Prediction, And Computing (Grad)
- **English**: Technical Writing, Advanced Technical Writing in Electrical Engineering

SKILLS

- **Systems**: Arduino, Nvidia TX Series, STM32, Franka Panda, Baxter, UR5e, DJI Manifold
- **Languages**: Python, C++/C, Matlab, Java
- **Tools**: Git, CMake, OpenCV, Tensorflow, Pytorch, Latex, OpenGL, ROS, Ubuntu
- **Hardware**: 3D Printing, Laser Cutting, Machining, Soldering